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Product-service systems for office furniture: barriers and opportunities on the European market

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Abstract

The environmental problems associated with office furniture consumption call for a more efficient use of office furniture that can decrease emissions and volume of waste. The author discusses the opportunities and barriers for the implementation of a PSS for office furniture. In addition, the paper identifies possibilities to overcome these barriers. The article is based on interviews with European office furniture producers, customers and experts that were confronted with a PSS scenario that was developed based on literature. The author concludes that the practical implementation of the developed PSS scenario will probably not lead to much success under the current market conditions and suggests how the PSS scenario can be adjusted to offer environmental benefits and to provide alternative business models to producers.
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1. Introduction

Product lifetime extension and remanufacturing strategies for used office furniture have been identified in the literature [1] as promising options to close material loops in the office furniture industry. One obstacle in regards to product lifetime extension strategies, such as repair and maintenance services, is that they diminish the producer's business opportunities to sell more new furniture. Some authors, for example [2], have proposed the concept of product-service systems (PSS) as a business model that can decouple the business success of companies from the amount of products sold.

The frequent replacement of office furniture in European companies contributes to the increasing production of solid waste that leads to the use of more landfill space. Germany produces approximately 700,000 tons (t) of office furniture waste per year, most of this waste going to landfill or incineration [3]. In addition to growing waste volumes that are expensive to handle, concerns about chemical inputs in furniture production were raised. These environmental challenges call for a more efficient use of office furniture, which could reduce waste volume as well as the quantity of hazardous raw materials used for furniture production. Contrary to environmental claims, the consumption trend moves towards shorter usage time of office furniture, since most companies do not "wear out" office furniture, but replace them for aesthetic reasons [4].

Some office furniture producers in Europe have adjusted their business strategies to environmental concerns and included environmental considerations in their design guidelines. The problem with environmentally sound and long-lived office furniture is that it is mainly offered in the upper price range. The sale of durable furniture also leads to a reduced sales volume, because customers do not need to replace the furniture as frequently.

One concept that may offer an opportunity to address environmental problems of office furniture in an

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economically feasible way is the concept of product-service system (PSS). PSSs are systems that try to complete or substitute traditional business models with a service offer that ideally leads to a reduction in environmental impacts. PSSs should "provide utility to consumers through the use of services instead of products" [5]. Manzini and Vezolli [2] state: "The Product Service System concept is a possible and promising business strategy potentially capable of helping achieve the leap which is needed to move to a more sustainable society". In theory, this concept looks promising both from an environmental as well as an economic point of view (for example [6]). However, a difficulty is that the switch from selling products to providing services does not automatically lead to a reduction in environmental impacts [7]. There seems to be a gap between the theoretical framework and the results of implemented service concepts.

For example, Vollmer [6] showed that existent leasing offers for office furniture in Germany are mainly financial instruments that neither address nor improve the environmental performance of furniture. On the other hand, Witte [1] highlighted the potential of service strategies as an important part of closed loop recycling in the office furniture industry that also offers new business opportunities for producers.

The existing literature is still missing an explanation for the absence of service concepts that take up the auspicious ideas about reuse and remanufacturing on the European office furniture market. This paper analyses the practical applicability of a PSS that addresses the environmental challenges of office furniture consumption through the following question: What are the barriers and opportunities for the implementation of product-service systems for office furniture on the European market and what are the possibilities to overcome these obstacles?

The article is based on 30 telephone interviews with office furniture manufacturers, customers and experts. Prior to the interviews, a PSS scenario for office furniture had been developed that served as a basis for discussion. The scenario evolved from an analysis of the relevant literature on service concepts for office furniture and the application of the general characteristics of PSS to the specific conditions in the office furniture industry. The goal of the interviews was to examine the attitudes and perceptions of practitioners in the office furniture business. The scenario was used as a starting point in order to discuss difficulties and opportunities for the implementation of PSSs in the context of office furniture. The results of the interviews were then used to adjust the primary PSS scenario. The PSS scenario describes a business approach where manufacturers rent office furniture to customers instead of selling it, which offers them the opportunity to remanufacture or refurbish the products between two renting periods.

2. Service concepts for office furniture: a proactive environmental strategy?

The most significant environmental impacts of office furniture are the aspects related to the production of raw materials and the disposal of old furniture [10]. Literature suggests two major strategies to tackle the environmental problems associated with office furniture consumption: one strategy is to design office furniture so that it is more suitable for material recycling [1]. The second one is to decrease office furniture consumption by e.g. prolonging furniture lifetime [6].

Witte [1] intensively examined different strategies for the implementation of closed loop recycling in the office furniture industry, including material recycling. A material recycling strategy involves the design of office furniture that facilitates easy dismantling and the separation of all materials that are then recycled. According to Witte [1], this strategy is the most suitable for cheap, mass furniture. Witte [1] pointed out that the most important aspect for material recycling from office furniture is the recycling costs. The sorting and disassembly of old furniture is expensive and it is questionable whether the revenue from the recycled materials can cover these costs. Witte [1] assessed the costs for the disassembly and transport of an office desk for material recycling and concluded that it was not economically feasible. The profitability of material recycling from office furniture is closely dependent on the volume of waste: the more office furniture is transported and disassembled together, the more profitable the process becomes [1]. In addition, it is extremely important that office furniture is designed for easy dismantling and that all materials are labeled to facilitate the separation. The prices for the recycled materials on the market also heavily influence whether or not it makes economic sense to recycle. Witte's results [1] indicated that material recycling for office furniture does not seem to be a promising strategy under the current conditions, since the recycling costs will most probably exceed the revenue from the recycled materials.

Several authors [1,6] have brought forward the idea of prolonging the lifetime of office furniture in order to reduce environmental impacts. Lifetime extension strategies for office furniture can include one or several of the following elements:

- *Design for durability*: In order to reduce the environmental impacts of office furniture consumption, the furniture should last for long periods. Environmentally proactive manufacturers such as Wilkhahn [13] are committed to product durability. The disadvantage of this strategy is that it undermines the business case for producers: if their products can be used over long periods, they will sell less furniture, at the end.

- *Maintenance and repair services*: Manufacturers can prolong the lifetime of their products by offering maintenance and repair services parallel to the sale of furniture. Witte [1] concludes that lifetime extension services for office chairs can be economically implemented. Furthermore, manufacturers can gain additional revenue from offering supplementary services to product sales [1].
- *Reuse of furniture parts*: This strategy is concerned with reuse of certain furniture parts for the production of new office furniture. The rationale behind this strategy is that certain parts of used furniture, such as metal pillars or desk support frameworks, are still valuable and should, therefore, be reused and not disposed of [1]. Witte [1] calculated that the utilization of old parts in new office furniture could lead to cost reductions of up to 35%. One problem for the implementation of this strategy is the development of a take back system that ensures producers a constant reflow of office furniture or furniture parts in good condition. Witte [1] suggested a model where producers offer a full-service-contract with the product's sale. Within this contract, producers offer maintenance and repair of their products and take back old elements to reintegrate them into the production of new furniture. The risk of this business model is that it could lead to lower sales figures.
- *Remanufacturing of used furniture*: The take back and remanufacturing of used office furniture can facilitate a second or third usage period. Office furniture is a simple product that in principle does not lose its function over many decades. Both the SYSKREIS project [3] and Witte [1] stated that a major fraction of production costs for office furniture is raw material costs. Therefore, it seems promising to remanufacture office furniture. According to a case study [1], office chair remanufacturing could be realized economically but difficulties remained in finding a market for second hand chairs.
- *Leasing or renting*: The idea to lease or rent out products instead of selling them evolved from the consideration that this strategy creates more incentives for producers to prolong the lifetime of their products, for example [8]. If the ownership of a product stays with the producer, s/he can directly benefit from the longer usage period. Vollmer [6] examined existing leasing concepts for office furniture and concluded that these offers are mainly financial instruments that do not support closed loop recycling. Goedkoop et al. [9] presented an interesting case study where office furniture was leased out to customers from a furniture pool that was included in the full-service-package of an office building. The SYSKREIS project [3] intended to develop and market systemic use-oriented office solutions. This research included the evaluation and adjustment of leasing models and other services as a basis for a more intensive use of office products. The SYSKREIS project [3] supported the idea of having an office furniture pool from which furniture is rented out to different customers. Unfortunately, the SYSKREIS project was not followed by a real market introduction of the developed concepts [10].

From an economic point of view, lifetime extension strategies seem to be a better option in order to close material loops in the office furniture industry compared to material recycling. Nevertheless, literature has also identified obstacles that hinder the implementation of these strategies. The main barrier is that lifetime extension strategies do not match up with traditional business models. The overall goal of product-oriented companies is to sell as many products as possible, contradicting the strategy of prolonging the usage phase of products. One idea to address these contradicting incentives is furniture leasing or renting. Existing leasing concepts for office furniture do not employ lifetime extension strategies. The office furniture market is still lacking a comprehensive service package that combines the different elements of the lifetime extension strategy to facilitate closed material loops.

3. The PSS scenario

The literature review revealed that there is no concept that facilitates the implementation of closed loop recycling in the furniture industry without destroying the business case for office furniture producers. A considerable obstacle is that all strategies for product lifetime extension lead to a reduced amount of furniture sold. On the other hand, strategies to implement material recycling of office furniture entail unreasonably high recycling costs. The task of this section is to present the PSS scenario that was developed to employ the remanufacturing and reuse strategy for used office furniture, since this idea has been identified as promising by several authors [1,3]. In addition, the PSS scenario should create incentives for manufacturers to invest in product lifetime extension strategies.

In general, product-oriented and use-oriented services are suitable PSS types for office furniture. A product-oriented service involves an additional service that is offered in combination with a product to improve the lifetime or utility of the product [10]. A use-oriented service exists when the actual product is owned by the service provider, who then sells the function of the product to the users [10]. Product-oriented services for office furniture are, for example, maintenance and repair services that are offered in combination with the sale of office furniture.

Maintenance and repair services for office furniture are quite often included in the product portfolio of manufacturers or retailers. They can serve as an additional income source for office furniture manufacturers or retailers and increase customer contacts. In addition, repair and maintenance services can prolong the use phase of office furniture and thereby reduce resource consumption and other environmental impacts. Nevertheless, these services are counter productive to the manufacturers' main business goal, to sell as much furniture as possible. It is also difficult to charge customers for maintenance and repair services if competition from cheap new office furniture is high. With use-oriented services, the use of the furniture is intensified through renting, leasing, sharing or pooling. Since office furniture is relatively heavy, it cannot be easily transported from one user to another. The use of office furniture by several customers during the same period does not seem to be very applicable in general, therefore pooling and sharing concepts were not more intensively assessed in this research. Nevertheless, there are some specific applications of office furniture where sharing might be an interesting service offer. For example, several companies, in one office building, can share meeting rooms in order to increase the usage time. Starting from these considerations a renting concept for office furniture seemed to be most suitable for creating incentives for producers to implement product lifetime extension strategies. A renting service would facilitate the regular take back of office furniture or parts for remanufacturing and reuse and was proposed by several authors [1,3] as an economically feasible option to close material loops in the office furniture industry.

The following PSS scenario therefore, puts emphasis on office furniture renting. However, it should be noted that other PSS concepts for office furniture are theoretically conceivable and could be developed as part of future research. In Europe and other parts of the world, several office furniture producers have already put the idea to rent or lease out office furniture into practice. In addition, researchers, e.g. the SYSKREIS project [3], have identified the opportunity to combine office furniture renting with reuse and remanufacturing concepts. The rationale of the following PSS scenario is to set up a complete model based on these already existing ideas in order to create a starting point for discussion with practitioners.

The development of the PSS scenario involved a clarification about which actor in the office furniture sector would be the most suitable for the role of the service provider. In general, either the furniture manufacturer or the retailer could take on the role of the service provider in the PSS scenario. The furniture retailer may expand its business and offer renting of furniture instead of just selling it. The company could take back and remanufacture furniture at the end of a renting period and rent it out again. The difficulty with this scenario is that it undermines the business case for the furniture manufacturer who will, as a result of the PSS, sell less furniture. In addition, the manufacturer has more know-how and more equipment for furniture processing and may therefore, be more qualified for remanufacturing. On the other hand, many furniture manufacturers choose to sell their products via retailers because they are not interested in organizing marketing and sales activities themselves. The outlined relations indicate that the PSS scenario should involve both retailers and manufacturers. One possibility would be for the manufacturer to become the service provider but still use the retailer as a link to the customer. This would mean that the manufacturer would rent the furniture to the customer directly and the company could employ the retailer as a kind of "sales department." The retailer would then organize the contract between the manufacturer and the customer, assisting the customer in choosing office furniture and informing the manufacturer about customer requirements. The manufacturer would concentrate on designing, producing and remanufacturing furniture. It is important that the same actor is responsible for the production and remanufacturing of the office furniture. Only this scenario creates incentives for the producer to design the furniture for easy remanufacture.

The following is the basic idea of the proposed PSS scenario: The furniture manufacturer offers products for renting. The customer rents the furniture and the service includes maintenance, repairing and up-grading by the manufacturer. At the end of the renting period the customer returns the furniture and receives remanufactured furniture if s/he wants to continue the renting contract. The old furniture is remanufactured and rented to another (or the same) customer. The process of remanufacturing involves the replacement and reuse of furniture elements and parts. The customer has the possibility to exchange furniture under the renting contract within fixed periods. Retailers could serve as a link between manufacturer and customer if the direct link proves to be difficult to establish and maintain. This service is close to modern facility management concepts where service providers rent out entire offices to companies including all technical equipment and office furniture, e.g. Regus [11]. Office furniture manufacturers could then provide the furniture service for several buildings in one area and pool a furniture stock for different customers. It should be noted that there are office furniture manufacturers and retailers on the European market who offer one or several parts of the proposed PSS scenario in their portfolio. However, none of the existing service systems for office furniture directly combine the furniture rental with a take back and remanufacturing strategy.

4. The PSS scenario and the reality

The proposed PSS scenario for office furniture has been intensively discussed in interviews with manufacturers, customers and experts in order to reveal how reasonable and practical the implementation of this theoretic framework would be. The goal was to collect information about real-life actors' perceptions of the PSS scenario in order to adjust and rethink the new business model.

4.1. Supply related factors

The European office furniture industry is currently experiencing stagnation and strong pressure from competition and customer demands. These developments have led to a concentration trend; meaning that many small companies had to shut down or were bought by bigger companies. These conditions seem to be ambivalent with regard to the implementation opportunities for the PSS scenario. On the one side, the market conditions call for new business ideas or innovations that could help companies escape from the destructive price war between competitors. On the other hand, the difficult situation does not leave much space for experimentation. The interviewed manufacturers and experts pointed to the strong competition from low cost countries, which could destroy the possibility to remanufacture furniture, since the remanufacturing of used furniture in western European countries would always be more expensive than the production of new furniture in low cost countries. Nevertheless, western European countries will most probably lose the price war against low cost countries, even if they rationalize production processes, just because of the difference in labor costs. Therefore, it may make more sense to offer something additional to the customer that low cost countries cannot offer. A renting service with the local presence of a service provider for repair, maintenance, reorganization, consultancy etc. may be one opportunity for national manufacturers to regain competitive advantage.

During the interviews, the remanufacturing process was recognized as the most critical part of the PSS scenario when it comes to economic feasibility. Interviewed manufacturers have argued that the remanufacturing of furniture would require more individualized work steps that could neither be standardized nor automated similar to the mass production of furniture and would result in much higher costs. It is not yet proven that this argument is true, since the PSS scenario will most probably lead to cost savings in other areas. For example, manufacturers will need to buy fewer raw materials if they reuse and remanufacture furniture. They will also save the labor and energy costs that are involved in the continuous production processes. It seems unreasonable to put everything but automated mass production down as unprofitable if no proper cost–benefit analysis has yet been conducted. In addition, literature [1,4] and interviewed manufacturers stated that raw material and labor are the major costs for furniture production. UEA [4], for example, states that the production value of furniture in the EU consists of 45% raw materials and semi-finished products and of 40% value added. It therefore, seems promising to reuse furniture or furniture parts. Manufacturers could save the costs for the most expensive production steps if they were able to remanufacture their products in a way that facilitates the 2nd or 3rd use phase. Another remark by one of the interviewed manufacturers was that the remanufacturing processes of the PSS scenario could hardly be organized and planned in an efficient manner, since it would be difficult to predict when and how much furniture will come back to the service facility. This statement points out the importance of a good contract for the renting service. The contract definitions should minimize the risk and unpredictability for the service provider. In addition, it may make more sense if a local service company that bundles the orders from different manufacturers does the remanufacturing in order to increase the volumes attained.

During the interviews, logistics emerged as another important factor with regard to the economic feasibility of the PSS scenario. The cost effectiveness of the PSS scenario significantly depends on the transport distance between the service provider and the customer, as well as the amount of furniture serviced by one provider. It seems only decentralized organization would make economic sense. Transport of office furniture seems to be too expensive to facilitate transport back to the production facility. Once the furniture has reached the customer it should be transported as little as possible for both environmental and economic reasons. Regional or local service providers seem to be the best solution in order to provide an economically feasible renting service. These providers should be responsible for repair, maintenance, customer consultancy, relocations, reorganizations and remanufacturing.

Another economic barrier that was pointed out during the interviews was the financial risk that office furniture manufacturers would have to take with the PSS scenario. The office furniture industry is characterized by many small and medium-sized enterprises (SMEs). SMEs would not be able to finance the PSS concept since they do not have enough financial resources. Manufacturers need to invest money in the production of the office furniture and a renting service requires much more time to cover the initial investment than product sale. In addition to the lack of financial resources, producers would have no experience and no competency to evaluate the financial risk of the renting scheme. There may be possibilities to adjust the PSS concept in a way that takes away the financial risk from

the manufacturers. One option may be to develop the renting contract in a way that minimizes the financial risk for the service provider, such as to introduce a minimum renting period. The alternative to organize the PSS in cooperation with a leasing company is problematic, because producers will have fewer incentives to produce durable furniture under these conditions (see Section 5).

Another obstacle that may constitute a general barrier for the implementation of new business models is that all organizations are by nature resistant to change. Several interviewees stated that a major factor that hinders the implementation of a PSS concept for office furniture is that producers do not want to switch from selling products to providing a service system. It seems that most managers in office furniture companies cannot imagine making money decoupled from the amount of furniture they sell. They seem to be afraid that renting could destroy their own business case.

4.2. Demand related factors

The interviewed experts pointed out that there is currently not much market demand for concepts such as the PSS scenario. An important finding was that customers make decisions based mainly on price comparisons. The PSS scenario could only become successful if it was cheaper than the furniture purchase. There seems to be no willingness to pay a premium price for furniture renting, even if it offered additional benefits. Interviews also indicated that office furniture might not be an ideal product for the implementation of a PSS. Office furniture is a simple product that does not require much maintenance or repair and has a considerably long usage time. The long usage time conflicts with the financial benefits of renting that only occur for short-term usage. The interviewed manufacturers and customers argued that renting of office furniture would always be more expensive in the long run. The technical simplicity of most office furniture would question the purpose of maintenance and repair services.

Besides price sensitivity, another identified barrier with regard to customer preferences was the importance of fashion and design. Extrinsic values of office furniture seem to be central in the purchasing decision. If furniture should be rented out for several years, there would be a big risk that the design would go out of fashion and no customer would want to rent it. In contrast, furniture rental offers the possibility to modify furniture frequently and adapt designs to current trends. It has to be noted that refurbishment and remanufacturing can only affect the surface of office furniture e.g. color, cushion or textile. Major characteristics of furniture such as shape, size or material cannot be changed through remanufacturing Even if the furniture for rent was regularly remanufactured, its design may appear old after 3–5 years, because office furniture design changes frequently. The importance of timeless design for rental furniture was frequently highlighted during the interviews. A problem may occur with employees' satisfaction and motivation if a company rents used furniture. These psychological barriers indicate that it may be better to reuse only parts of old furniture for the assembly of new furniture, so that customers cannot feel the difference between the old and the new furniture.

Another important factor identified during the interviews was the representative function of office furniture. Companies seem to have a negative perception with regard to the use of rented office furniture, because it may damage the company's reputation. The argument that customers always need fashionable office furniture is a major marketing argument that defends the frequent purchase of new office furniture. Further research is required in order to clarify the role of fashion and design in the value proposition of office furniture.

4.3. Environmental factors

The most significant environmental impacts of office furniture are the aspects related to the production of raw materials and the disposal of old furniture [12]. The proposed PSS scenario addresses both of these impacts. Renting of office furniture could lead to a reduced use of raw materials because it could prolong the usage time of furniture. In addition, the PSS scenario could lead to a reduction of disposed furniture. Under the current legislative framework, producers of office furniture are not required to organize the take back of their products. There are only a couple of innovative manufacturers, for example Wilkhahn [13], which have implemented eco-design guidelines and voluntarily offer to take back their products. The interviewed producers did not expect an implementation of Extended Producer Responsibility (EPR) for furniture in the near future.

The current waste management practices for old office furniture are not very promising, since most furniture is disposed on landfills or incinerated, for example [4]. There are no indications that the material recycling of office furniture has increased significantly during the last 10 years. Witte [1], who has conducted research on the possibilities of material recycling from office furniture waste, concluded that it was currently not economically feasible. The problem is that there is no market for recycled materials and no interest from the customers' side to pay for material recycling. Under the current legislative conditions in the EU it is cheaper and easier to burn or dispose old furniture. These legislative conditions can be interpreted as a barrier for the implementation of the PSS scenario. If manufacturers have no legislative pressure to rethink the way furniture waste is currently handled, they have no

incentive to implement concepts for closed loop recycling or product lifetime extension.

The interviewed manufacturers were not very concerned about the environmental impacts of their products beyond the production emissions. Their environmental efforts seem to focus on the performance of their production facilities. Customers' interest in environmental performance of office furniture producers was limited to the requirement for an ISO 14001 certified environmental management system. These results are not very promising for the implementation of the PSS scenario. Neither interviewed manufacturers nor customers seem to have an environmental motivation to switch to a PSS for office furniture. Therefore, the environmental benefits of the PSS scenario will probably not be useful for the marketing of the concept.

4.4. Opportunities for a practical application

It can be concluded from the interviews that there are some applications that are more preferable for the implementation of the PSS scenario than others. During the discussions with producers, it was mentioned that high quality furniture and furniture with technical features is generally more suitable than simple or low quality furniture. It seems logical that furniture with a higher value is better to remanufacture and reuse. The more technical features the furniture has, the more reasons exist for offering a service for repair and maintenance. Short-term projects seem to be the ideal application of the PSS scenario, but the size of this section of the market is questionable.

It was also mentioned during the interviews that back office might be the market segment that is more appropriate for the application of renting furniture. This seems reasonable, because in back office the visual appearance of furniture is less important. However, back office is the market segment that normally purchases mass furniture of low quality. Intensive market research will be needed in order to define a target market segment and a certain quality of the furniture for the renting concept. Renting of office furniture also seems to be more applicable for big companies, because the liquidity benefits are more significant and the efforts of the service provider are more profitable.

The same logic applies to the concept that renting systems for office furniture should preferably be developed in big cities with a lot of potential customers in a relatively small geographic area. Another possibility that was put forward by many interviewees was the integration of the PSS scenario in an all-inclusive renting concept for offices, which already exists in several cities around the world, e.g. [11]. These service packages offer the possibility for customers to rent offices that are fully equipped with furniture, carpets, IT equipment, communication technology, lightning, etc.

The interviews with experts who do research in the field of work organization revealed that the PSS scenario seems to have more opportunities for successful implementation in companies that are already reorganizing their work processes according to future trends. New Work is a catch phrase that is used in connection with future developments of office work. New Work is the application of new working methods as an adaptation to international trends attendant to globalization [14]. One effect of New Work will be that individual workplaces will become less important for employees, which could increase their acceptance of rented office furniture. In addition, short-term project group work seems to become more usual, which requires a frequent reorganization of office interiors. It will perhaps be easier to organize these internal reorganizations from a big rental furniture pool instead of only having access to a limited amount of company owned furniture. Another consideration with regard to New Work is that these future developments could lead to a significant decrease of furniture consumption by many companies, since home offices and other flexible working concepts will increase. In addition, New Work trends call for more flexibility, which cannot be easily fulfilled by the traditional sales concept without diminishing the income flow for furniture manufacturers. Office furniture manufacturers therefore have a good reason to decouple their revenues from the amount of consumed furniture and instead get involved in different services for offices. It may consequently be a good idea to combine the realization of a furniture rental concept with the implementation of working concepts that follow New Work. The renting concept could profit from the positive and more fashionable image of New Work that a stand-alone presentation of the PSS scenario would lack. The increasing popularity of facility management concepts also supports the idea that companies outsource the responsibility for their office interiors. Interior space planning is one possibility for office furniture manufacturers or retailers to enter a new business.

5. Adjustment of the PSS scenario

The interviews revealed three main problems with regard to the practical application of the PSS scenario. Firstly, it seemed not to be economically feasible to transport the furniture back to a central production facility for remanufacturing. Secondly, the service provider would have to take a very high financial risk when renting out his products. Thirdly, the original PSS scenario was too simple to attract customers.

The ownership of the furniture should stay with the producer in order to create incentives for product development and design that is focused on product

lifetime extension. Nevertheless, it would be feasible that, for example, retailers, facility management or leasing companies could take on the role of the service provider instead. These companies would then only buy from those manufacturers that have durable products that are good for remanufacturing as well as repair. One could argue that this business model would also create incentives for manufacturers to design durable furniture, because it would create competitive advantage. But at the end, producers who sell furniture have always had one major business goal: to sell more furniture. In a concept where manufacturers sell to someone who rents out their product, the incentive would be to offer a product that is more durable than the average product but which is worn down fast enough to secure a sales market for new products. It would lead to a sub optimal product development from a resource productivity point of view.

Therefore, the question that has to be answered now is not: *should* the manufacturer be a service provider at all, but *how* can the manufacturer organize the remanufacturing of the furniture outside the central production facility? One possibility would be that the manufacturer scales down the central production facility and builds up several decentralized service facilities close to the company's most important customer centers. It should also be mentioned that these considerations show that the PSS scenario cannot replace office furniture sales totally. Decentralized service facilities can only be run in areas where the number of customers facilitates efficient operations. This is to say that customers in geographically remote locations cannot be target customers for furniture renting since the logistic costs would be unreasonably high. In regions with high density of office workplaces small service units could serve local customers and provide all necessary assistance around the furniture rental such as maintenance, repair, remanufacturing etc. A manufacturer that focuses on furniture renting will only need a small central production facility, which would have irregular production cycles for new furniture, spare parts as well as parts to up-grade and supplement furniture. In contrast, the product development and product design units probably need to expand in order to support the renting concept. The marketing efforts, almost certainly, need to be increased significantly, at least in the beginning when the whole concept is launched.

Another possibility would be that manufacturers would enter partnerships with local service companies that could carry out different kinds of services supporting furniture renting. In this model, the manufacturer would rent out products and then find partners that take over the practical accomplishment of repair, maintenance, remanufacturing, etc. The manufacturer would provide his partners with know-how and spare parts. Possible partners for this model could be, for example, retailers.

The financial risk that every service provider would take when renting out office furniture is a big problem and it seems difficult to find a sound solution. The service provider will always run the risk that the furniture will not be rented out long enough to cover the company's investments. On the other hand, in the traditional business model it was the customer who was taking the same risk when purchasing office furniture. What if the customer had to decrease their workforce after two years and did not need the furniture any more? One suggestion discussed during the interviews was to formulate the renting contract in a way that minimizes the financial risk for the service provider. For example, the contract could include a minimum renting period of 5 years. During these first 5 years the customer cannot return the furniture. If the customer wants to cancel the contract, they have to buy the furniture from the service provider for a fixed amount. This amount will cover the investment costs of the manufacturer and additionally give some kind of profit. The renting rates for the furniture will be composed of two parts. One part, called service rate, will cover all services that are included in the contract (for example, annual cleaning, repairing, remanufacturing every 5 years etc). The service rate will stay constant over the entire contract period. The second part will be the renting rate, which is what the customer pays for using the furniture. The sum of all renting rates of the first 5 years should approximately cover the investment of the service provider. After the first 5 years, the renting rate will decrease in fixed periods in order to create an incentive for customers to keep their furniture for as long as possible. However, the sum of the service rate and the renting rate should still create enough profit for the service provider so that the provider has no incentive to convince the customer to replace his furniture with newer models.

The price sensitivity of customers and their low interest in environmental improvements of office furniture revealed that the service provider needs to offer additional services in combination with furniture rental in order to catch the attention of clients, e.g. space planning, furniture inventory analysis, consultancy for work organization, etc. These additional services could stabilize the relationship with the customer, so that the customer would lose more than just office furniture when quitting the contract. Customer retention would possibly be higher this way.

6. Discussion

The previous analysis has discussed the results of this research regarding the applicability of the PSS scenario in the office furniture industry. This section provides an analysis in a wider context by discussing how the results

can be interpreted in regards to the PSS idea in general as an instrument for the development towards a sustainable society. The evaluation of the PSS scenario for office furniture has shown that there are many obstacles in the industry that could hinder a successful implementation of the concept. A question that has to be raised is whether these barriers are branch specific or whether some major obstacles exist in our economy and society that obstruct the shift towards a service society.

The interviews pointed out a phenomenon that seems to constitute a barrier to the implementation of all new ideas. Humans are resistant to change. We are afraid to leave traditional business models and have difficulties in rethinking the fundamentals of our economy. During the interviews, the impression evolved that many people were strongly loyal to certain structures and could not imagine changing these structures. A new business model needs to propose explicit benefits compared to traditional business models in order to gain attention. It seems that people are only willing to take the risk of leaving traditional paths if they expect a significant profit from doing so. A general problem of environmentally innovative PSSs may be that their benefits are not explicit to most business people. In an economy were many environmental impacts are still not accounted for in market prices, company leaders do not logically consider the environment when they make decisions about products and system design [15].

The influence of fashion and the trends of office furniture was another barrier that was identified during the interviews. Renting of office furniture could be difficult because people are not willing to rent something that is out of fashion. This result may also illustrate a general problem of our society that hinders sustainability efforts. We are all heavily influenced by fashion and cannot make buying decisions totally independently and rationally. At the end, fashion seems to be nothing more than a marketing idea of companies. In the case of office furniture especially, it seems absolutely irrational that fashion seems to have such a big influence. How much revenue does the replacement of office furniture every 3–5 years create? This revenue is hard to measure, but nobody seems to question the value of up-to-date furniture. If a company buys a machine for the production facility, the managers check in detail how this machine will contribute to the company's profitability. One explanation for this behavior with regard to the purchasing decision of office furniture could be that furniture costs are quite low compared to other investments a company makes. However, in regards to the significant changes that are needed in our society to solve the environmental problems we are facing, every improvement counts no matter how small it is.

One result of this research may be that it is problematic that our society does not pay attention to small issues. It is definitely clear that the waste problem resulting from office furniture consumption is not considered as urgent as radioactive waste, for example. In addition, it is understandable that companies do not invest much energy and time in the purchasing decisions of office furniture if it counts for less than 1% of the costs of an office building. Nonetheless, many companies are struggling for survival and should take every chance to increase their profitability. Changing a few major issues will not solve the environmental problems of planet earth and the struggle for resources. Our society and economy is a puzzle of innumerable pieces and we have to change our behavior in every little piece in order to see a different picture.

With regard to the value of this investigation for overall PSS related research, it can be concluded that it seems there are some product characteristics that support the idea of PSS better than others. Many interviewees mentioned that office furniture was not a suitable product group for the implementation of the PSS concept. This judgment should not be overestimated, but it may be helpful to name some product characteristics that seem to be advantageous for a successful implementation of PSSs. The results of the interviews support Tischner's [16] findings that products with one or more of the following characteristics seem to be especially suitable for the PSS business model:

- Expensive products;
- Technically advanced products that require maintenance and repair;
- Products that are easy to transport;
- Products that are used infrequently by customers;
- Products that are not heavily influenced by fashion or trends.

It is obvious that none of these characteristics satisfactorily apply to office furniture. The list is not exhaustive, but provides some indications that were found during this study. It may be helpful if future research would examine the role of these product characteristics, in more detail, in order to give business people advice on selecting products for the development of PSS concepts. Since the implementation of the PSS concept faces many barriers in reality, it might be good, in the beginning, to focus on those product groups that have the most optimal characteristics.

7. Conclusions and recommendations

The objective of this paper was to identify and assess barriers and opportunities for the implementation of the PSS concept for office furniture on the EU market. Another goal was to deliver some recommendations for

how these obstacles could be overcome. The main barriers identified in this research are the following:

- *Financial risk for the service provider*: The service provider is taking a considerably high financial risk when renting out office furniture. The main risk is that the provider does not find customers for a sufficiently long time and cannot cover investments.
- *Market conditions*: The market is very competitive, at the moment, and producers compete mainly based on prices. There seems to be a lack of willingness from the customers' side to pay a premium price for environmentally superior products or additional service offers.
- *No legislative pressure for and no interest in environmental improvements*: The legislation in the EU does not require manufacturers to take back their products. Manufacturers do not seem to think that EPR for furniture will be implemented in the near future. The interviewed manufacturers did not show much interest in the environmental benefits of the PSS scenario.
- *Characteristics of office furniture*: Customers normally use office furniture over long periods (on average 12 years) and this does not seem to support the idea of renting. In addition, office furniture is a simple product that does not need much maintenance or repair.
- *Resistance to change*: Manufacturers are used to selling furniture and it would require significant mental and organizational changes to switch to a renting concept. Customers are also used to buying office furniture and their purchasing departments may be resistant to abandon traditional purchasing practices.
- *Importance of fashion and design*: The research revealed that fashion and design are important characteristics of office furniture for customers. The idea to use office furniture even longer than already practiced seems to stand in contradiction to customers' needs for up-to-date furniture design. Resistance to change and the influence of fashion have been identified as two factors that may be barriers to sustainable development in general.

The interview results did not deliver an absolute clear picture on the practical applicability of the PSS scenario. Nevertheless, some general conclusions can be drawn that may help the further development of PSSs for office furniture:

- *Profitability*: It remains unknown whether the PSS scenario is economically feasible or not, since it was not the objective of this research to calculate the profitability. The interviewed manufacturers were quite critical about the profitability of office furniture renting and especially concerned about the costs for remanufacturing. Experts and literature delivered a more positive picture on the profitability of services for office furniture. It can be concluded that the profitability of the PSS scenario strongly depends on the transport distance between customers and service providers as well as the number of furniture serviced by one service provider. Further research is needed in regards to the profitability of the PSS concept. A useful next step may be to conduct a case study and calculate the profitability of this application.
- *Organization*: The PSS scenario should be organized with regional or local service providers. A central organization of the concept makes the logistic costs unreasonably high. The results show that the PSS scenario is certainly not applicable in regions where customers are geographically wide spread. Further research is required in order to define the maximum transport distance that secures profitability for the service provider. The overall goal of the PSS concept for office furniture should be to increase the usage time of each customer as much as possible. Too short renting periods would lead to increased transportation, which could use up the environmental and economic benefits of the concept.
- *Market demand*: The market demand for a PSS for office furniture and the customers' needs remains uncertain. It seems that every concept that offers significant cost advantages to customers could become successful. There are doubts whether the characteristics of office furniture can support the implementation of a service concept. Two critical features are the long usage time and the technical simplicity of office furniture. As previously mentioned, there are indications that the extrinsic values of office furniture, such as fashion and design as well as representative functions, influence the purchasing decision significantly and therefore constitute a barrier to furniture renting.
- *Service package*: The PSS concept should create strong incentives for customers to rent furniture for as long as possible. Since a long usage time normally creates economic incentives to buy a product, it is important that the service provider includes additional services in the rental package. These additional services may help to create strong relations between customers and producers. The additional services could also help improve the attractiveness of the PSS scenario.
- *Opportunities*: The implementation of the proposed PSS scenario certainly has some potential to improve the situation for manufactures, customers and the environment. Since the office furniture industry in the EU is presently in crisis, service concepts may offer producers a chance to escape from the price war with low cost countries. The

development of office work in the future could facilitate the implementation of the PSS concept for furniture. The philosophy of New Work and the PSS scenario seem to be in accord with one another.

- *Renting contract*: It can be concluded that a good contract is of high importance for the success of the PSS scenario. The setting of the contract conditions for the renting concept could minimize the financial risk and the unpredictability for the service provider.
- *Environmental benefits*: From an environmental point of view, the adjusted PSS scenario offers the advantage to address the most significant environmental impacts of office furniture: raw material production and furniture disposal. There are no indications that the problem of waste from office furniture will be solved in the near future. Material recycling is still not a common practice in the industry and this research concludes that there is no evidence to show that the industry will move toward material recycling. Manufacturers have no legislative pressure and no economic incentives to support material recycling of their products. The PSS scenario may be an opportunity to address the waste problem of office furniture in a more constructive way.

It can be concluded that the practical implementation of the PSS scenario suggested in this paper will probably not lead to much success under the current market conditions. Nevertheless, since the adjusted PSS scenario offers considerable environmental benefits and has the potential to provide alternative business ideas to producers, further research should be conducted. A next step could be to more thoroughly assess the market demand and the profitability of the concept. Customers' needs and patterns of office furniture use should be examined in more detail. The following questions could, among others, be asked: how important is fashion and design in the purchasing decision? How important are the extrinsic values of office furniture for employees' motivation and satisfaction? What are employees' attitudes towards office furniture renting? In addition, the research focus should be set on the following market segments or business ideas, since they seem to offer most potential for the application of a renting concept for office furniture:

- Large companies with many office workplaces;
- High quality furniture;
- Back office;
- All-inclusive renting of offices;
- Companies that have a lot of short-term project work.

In regards to the environmental problems of office furniture, it can be concluded that legislators should rethink the implementation of EPR for furniture, since it would promote the implementation of lifetime extension strategies in the furniture industry.

Furthermore, this study concludes that the product characteristics of a product group significantly influence the success of a PSS. Further research should therefore, be conducted on the role of certain product characteristics for the successful implementation of PSSs. This research could be conducted by analyzing success stories of implemented PSS concepts. Another approach would be to theoretically define product criteria, select products that fulfill these criteria and test whether or not the criteria lead to a successful PSS in a case study. The outcome of this research could provide guidelines for business managers that assist in the selection of product groups for successful PSS development.

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References

[1] Witte K. Büromöbel für die Zukunft: wandlungsfähig – langlebig – kreislauftauglich. [Office furniture for the future: versatile – durable – recyclable]. Aachen: Shaker Verlag; 2000.

[2] Manzini E, Vezolli C. Product-service systems and sustainability–opportunities for sustainable solutions. United Nations Environment Programme. Division of Technology Industry and Economics (DTIE), Production and Consumption Branch; 2001. [Online] Available from: Available: http://www.uneptie.org/pc/sustain/reports/pss/pss-imp-7.pdf.

[3] Nolte R, Wapler M. Verbundprojekt SYSKREIS –Nutzenoptimierte Systemlösungen zur Förderung der Kreislaufwirtschaft am Beispiel Büro – Abschlussbericht in Kooperation mit dem Institut für Werkzeugmaschinen und Fabrikbetrieb, TU Berlin [Network Project SYSKREIS – Utility optimized system solutions for the promotion of closed loop recycling considering as example offices – final report in cooperation with the Institute for Machine Tools and Processing Operations]. WerkstattBericht Nr. 48. Institut für Zukunftsstudien und Technologiebewertung [Institute for Futures Studies and Technology Assessment] im Auftrag des Bundesministeriums für Bildung, Wissenschaft, Forschung und Technologie, Berlin; 2001.

[4] UEA – Federation of European Furniture manufacturers. The furniture waste site; 2004. [Online] Available from: http://www.ueanet.com/furniturewaste/frame4.htm.

[5] Mont O. Clarifying the concept of product-service system. Journal of Cleaner Production 2002;10(3):237–45.

[6] Vollmer A. Büromöbelleasing – Chancen eines öko-effizienten Dienstleistungskonzeptes. Theoretische Grundlagen und erste praktische Ansätze [Office furniture leasing – opportunities of

an eco-efficient service concept. Theoretical background and first practical approaches]. Diplomarbeit an der Fachhochschule für Technik und Wirtschaft Berlin; 1999.

[7] Mont O. Product-service systems: Panacea or myth? Doctoral Dissertation. International Institute for Industrial Environmental Economics, Lund University; 2004.

[8] Schrader U. Konsumentenakzeptanz eigentumsersetzender Dienstleistungen: Konzeption und empirische Analyse. [Consumers' acceptance for services which substitute property]. Markt und Konsum, Band 10. Peter Lang Verlag; 2001.

[9] Goedkoop M, van Halen C, te Riele H, Rommens P. Product-service systems, ecological and economic basics. The Netherlands: Pre consultants; 1999. [Online] Available from: http://www.pre.nl/pss/default.htm.

[10] Nickel R, Verkuijl M, Tischner U. SusProNet Report – Product Service Systems to Offices – State of the Art Study. Econcept, Cologne, Germany; 2003. [Online] Available from: www.suspro-net.org.

[11] Regus Business Centres: The global leader in serviced office space with executive office suites, meeting rooms and virtual office solutions; 2004. [Online] Available from: http://www.regus.com/.

[12] Hopfenbeck W, Jasch C. Ökodesign – Umweltorientierte Produktpolitik. [Eco design – environmentally-oriented product policy]. Landsberg/Lech: Verlag Moderne Industrie; 1995.

[13] Wilkhahn. Environmental Statement 2001. Bad Münder, 2002.

[14] Hungenberg W. New Work – Bürozukunft heute. Neue Arbeitsmethoden, Organisationsformen und Bürokonzepte. [New Work – Office future today. New working methodologies, organizational structures and office concepts]. Deutsches Büromöbel Forum im Verband Büro-, Sitz- und Objektmöbel e. V; 2004.

[15] Mont O, Lindhqvist T. The role of public policy in advancement of product-service systems. Journal of Cleaner Production 2003;11(8):905–14.

[16] Tischner U, Verkuijl M, Tukker A. Product-service systems: best practice document: SusProNet; 2002.